AN LLM TUTORIAL | MTSAMPLES + SYNTHEA + GPT-4O-MINI

Comparing LLM Prompting Methods

on Healthcare Text and Structured Data

Zero-shot, few-shot, chain-of-thought, and tree-of-thoughts prompting, tested on the same goal across two very different kinds of healthcare data: free-text clinical reports and structured patient records.

SLIDES AND ANALYSIS DRAFTED WITH AI ASSISTANCE | CONTENT AND RESULTS REVIEWED BY THE AUTHOR

MOTIVATION

Why Compare Prompting Methods?

Same goal, four ways

Every method gets the same task and the same output format. Only the prompt changes.

Two different data shapes

Free text and structured records stress prompting differently. Comparing both shows what really helps.

Honest evaluation

Every result is checked against real ground truth. Formatting artifacts get caught and corrected, not hidden.

Prompting technique is often chosen by habit. This tutorial tests that habit: does extra reasoning actually help, or just add complexity?

SETUP

Two Datasets, One Common Goal

MTSamples

- ~5,000 clinical reports, 40 specialties
- Goal: classify medical specialty
- Ground truth: the label already in the data
- Eval set: 49 records, 7 specialties

Synthea

- Simulated patients, full condition + medication history
- Goal: classify opioid-misuse risk (Low / High)
- Ground truth: later "Drug overdose" condition, used as a proxy
- Eval set: 50 patients, 25 High + 25 Low (oversampled)

GROUND TRUTH

How We Know What's Correct

NOTE

MTSamples ground truth is a real label already in the data. Synthea ground truth is a proxy: a later "Drug overdose" condition marks a patient High risk — bounded by what the simulator generated, not a validated clinical score.

Each Synthea patient's history is cut off before their overdose date, so the model never sees the outcome it's predicting.

METHODS

Four Prompting Methods, Same Question

1. Zero-shot

Just an instruction, no examples. Tests raw model ability.

2. Few-shot

1-3 example pairs shown first, from a separate pool.

3. Chain-of-thought (CoT)

Reason step by step internally, then output only the label.

4. Tree-of-thoughts (ToT)

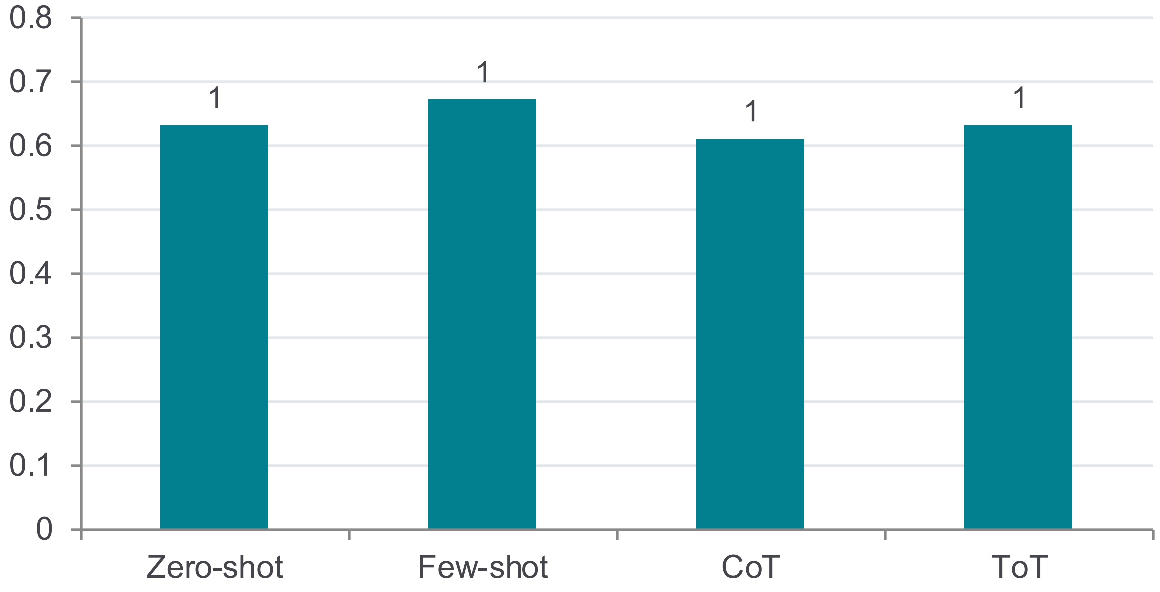
Weigh three interpretations internally, pick the best one.

An Example: Chain-of-Thought on Synthea

```
Given this patient's condition and medication history, classify
their opioid-misuse risk as Low or High.

First, silently reason step by step: (1) list the conditions in
chronological order, (2) list the medications in chronological order,
(3) note any pattern of escalation, switching, or discontinuation in
medication type/potency over time, (4) classify risk based on that
pattern. Then output ONLY the final JSON (do not show your reasoning
steps in the output).

Respond ONLY with JSON: {"label": "Low" or "High",
"probability_high": <float 0.0-1.0>, "reason": "<one sentence>"}

[ patient's condition + medication timeline inserted here ]
```

All four methods share this same JSON schema — only the reasoning instructions in the middle change.

Specialty Classification Accuracy

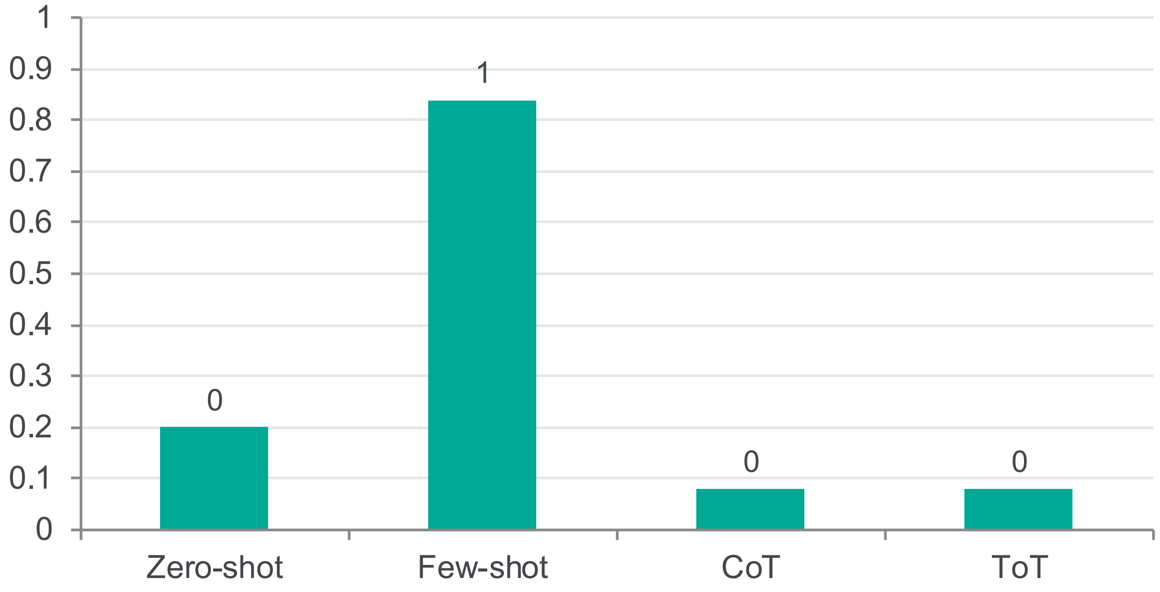


FINDING

Compound specialty names, like "Cardiovascular / Pulmonary", were sometimes answered as just one half. That was correct, but scored as wrong.

After fixing this, all four methods land close together (0.61-0.67). The earlier gap was mostly a formatting issue, not a reasoning difference.

Opioid-Risk Recall by Method



FINDING

CoT and ToT almost never predicted High risk (recall 0.08), though always correct when they did (precision 1.00).

Few-shot performed best (recall 0.84) — concrete examples anchored the model better than reasoning did.

CoT and ToT gave nearly identical results, so the extra step in ToT added no real benefit here.

CAVEATS

What These Results Don't Show

- Synthea ground truth is a proxy, not a validated risk score.
- High-risk patients were oversampled for evaluation (25 of 50, vs. a natural 4.4% rate).
- Eval sets are small (49 and 50 records) to keep cost and runtime low.
- Few-shot examples had zero overlap with the eval records.

NEXT STEPS

Ideas for Improvement

Fix labels before scoring

Normalize predictions against the full specialty list so partial matches aren't penalized.

Add AUROC / AUPRC

Score the model's confidence directly, not just its 0.5-threshold label.

Try an embeddings classifier

Train a simple classifier on embeddings as a non-prompting baseline.

A larger, more balanced eval set would help confirm whether the CoT/ToT caution effect holds up.

WHAT WE LEARNED

More Reasoning Isn't Always Better

Once a labeling issue was fixed, all four methods scored about the same on MTSamples. On Synthea, CoT and ToT made the model too cautious, missing most true high-risk patients — few-shot was the most reliable anchor on both tasks.